

⚡ End-to-End Data Pipeline & LLM Caching Architecture

1 News Ingestion

RSS / Atom feeds or search APIs retrieved via decoupled NewsSource adapters with topic pre-filtering.

```
DuckDB: news:rss [TTL]
```

2 AI Extraction

Pydantic-AI Agent extracts claims & attacks into typed Pydantic models. Strictly forbids URL hallucination.

```
DuckDB: framework:key
```

3 Formal Inference

Mathematical evaluation of Preferred Extensions according to Dung (1995) via backtracking or SAT solvers.

```
DuckDB: extensions:naive
```

4 Topology & Metrics

Graph-theoretic scores (acceptance rate), dilemma axes ($A \leftrightarrow B$), and in/out degrees.

```
In-Memory (Pure Functions)
```

5 AI Synthesis & Export

Second Pydantic-AI Agent synthesizes perspectives into concise theses. Automated timestamped export.

```
output/YYYYMMDD_*.md
```

📰 Real-World Pipeline Scenario: "Munich Olympic Bid"

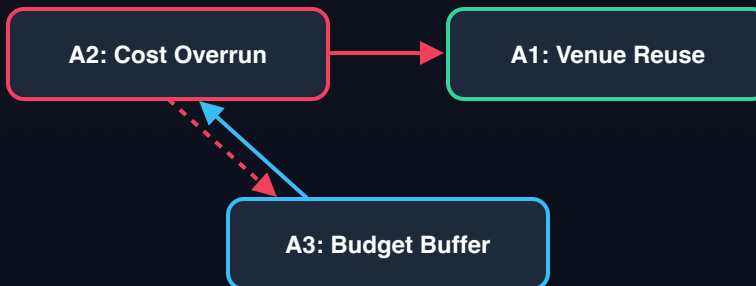
1. EXTRACTED ARGUMENTS (A)

A1 (Economy): Reusing existing venues revitalizes sports infrastructure and local economy.

A2 (Fiscal Risk): Cost overruns impose unpredictable burdens on taxpayers.

A3 (Safeguard): Substantial reserve buffers and strict transparency rules mitigate fiscal risks.

2. ARGUMENT GRAPH & ATTACKS (R)



A2 attacks A1 | A3 defends A1 against A2 | $A2 \leftrightarrow A3$ form a dilemma axis

3. COMPUTED PERSPECTIVES

Perspective 1 (Pro-Bid): {A1, A3}

AI Thesis: Sustainable opportunity for regional sports and economy under controlled fiscal risk.

Perspective 2 (Anti-Bid): {A2}

AI Thesis: Uncontrollable expenditure risks outweigh any potential long-term benefits.

🤖 Dual-Stage LLM Integration

Pydantic-AI Powered: Stage 1 extracts unstructured news into typed `ArgumentationFramework` graphs. Stage 2 takes mathematically proven extensions and generates natural-language meta-theses.

```
uv run clashpy --solver naive
```

📊 Structural Roles

Mathematical classification by acceptance score across computed extensions:

Core Consensus (Score = 1.0)

Contested ($0 < \text{Score} < 1$)

Excluded / Rejected (Score = 0)

🔒 Zero-Leakage Secrets

Fully automated API key resolution without hardcoding:

- `os.environ`
- Local `.env` file
- Encrypted OS Keyring (`db.syst.datahub`)